

Total No. of Questions : 6]

SEAT No. :

**P 1259**

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**APR - 18/T.E/Insem - 142**

**T.E. (Computer Engineering)**

**Embedded System & Internet of Things**

**(Semester - II) (2015 Pattern)**

*Time : 1 Hours]*

*[Max. Marks :30*

*Instructions to the candidates:*

- 1) Answer any three questions Q1. or Q.2, Q3 or Q4 and Q5 or Q6.*
- 2) Assume Suitable data wherever necessary*
- 3) Figures to the right indicate full marks.*
- 4) Draw neat & labelled diagram wherever necessary.*

- Q1)** a) Can an electronic tablet be listed as an embedded system? Justify your answer. **[4]**
- b) List any four advanced features of ARM core. **[2]**
- c) List various levels of IoT system and explain Level 1 IoT system with diagram. **[4]**

OR

- Q2)** a) Define release time, scheduling time, completion time and run time. **[4]**
- b) List various IoT communication models **[2]**
- c) List different IoT enabling technologies which play a key-role and explain any of them. **[4]**
- OR
- Q3)** a) Explain purpose and requirements specifications step of IoT system design methodology, consider smart IoT - based home automation system as an example. **[5]**
- b) Explain information model specification step of IoT system design methodology, consider smart IoT - based home automation system as an example. **[5]**

OR

**P.T.O**

**Q4) a)** With the help of diagram list and briefly explain the steps involved in the IoT system design methodology. **[5]**

b) Explain process model specification step of IoT system design methodology, consider smart IoT- based home automation system as an example. **[5]**

**Q5) a)** Draw and explain the four pillars of IoT paradigms. **[4]**

b) Write a python program for blinking LED Using Raspberry pi board **[4]**

c) Explain the concept of the Horizontal and verticals in IoT. **[2]**

OR

**Q6) a)** Write a short note on 5A and 3I characteristics of IoT. **[3]**

b) List and explain various features of Raspberry Pi board (Model B Revision 2). **[4]**

c) List any six IoT devices available in market. **[3]**

